A genus-two cover of your binary elliptic curve

Explicit construction and native subgroup transfer

14 October 2026 — Run `run-20261014T1442Z`

For either $a = 0$ or $a = 1$, over the same field $k = \mathbb{F}_{2^{83}}$, take

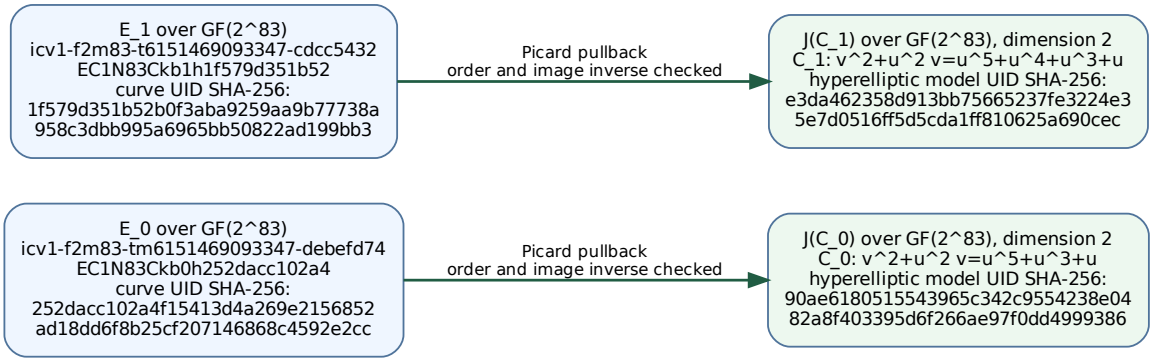
$$E_a : \; y^2 + xy = x^3 + ax^2 + 1, \qquad \boxed{C_a : \; v^2 + u^2 v = u^5 + au^4 + u^3 + u.}$$

The smooth projective curve C_a has genus two and a separable degree-three map

$$\boxed{\pi(u, v) = \left(\frac{u^3}{(u+1)^2}, \frac{u(u+1)v + u^2 + u + 1}{(u+1)^3} \right).}$$

At poles, use the projective extension: source infinity and the two geometric points over $u = 1$ map to O; $(0, 0)$ maps to $(0, 1)$.

The group law is on the abelian surface $J(C_a)$, using the homomorphism $\pi^* : E_a \to J(C_a)$. Both curves have dimension one; $J(C_a)$ has dimension two. All points and maps stay over $\mathbb{F}_{2^{83}}$. This is a same-field construction.



The diagram gives complete source identities and target-model SHA-256 digests (concatenate each digest's two lines). Target identifiers use `urn:hyperelliptic-model:sha256`; they are curve-model identifiers, not EC1 subgroup identities. The cover equation and exact fields determine the models.

Observed: both pinned subgroups transfer, their generators keep their orders, the public $Q = 37P$ relation is preserved, and the image inverse recovers both P and Q. A faster destination solver has not been established.

Why the construction works

Genus and smoothness. The affine equation has v derivative u^2; at the only possible singular point $(0,0)$ its u derivative is one. At infinity, $t = 1/u$, $z = v/u^3$ give

$$z^2 + tz = t + at^2 + t^3 + t^5,$$

smooth at $(t,z) = (0,0)$, with one rational point at infinity. On $u \neq 0$, set $w = v/u^2$. Then

$$w^2 + w = u + a + u^{-1} + u^{-3}.$$

The two Artin–Schreier poles have odd orders $3, 1$, so the cover is geometrically connected and $2g + 2 = (3+1) + (1+1) = 6$: $g = 2$.

Equation and degree. Normalize E by $s = (y+1)/x$, giving $s^2 + s = x + a + 1/x$. Write $R = u^3/(u+1)^2$ and $s = w + 1/(u+1)$. The identities

$$R = u + \frac{u}{(u+1)^2}, \qquad \frac{1}{R} = \frac{1}{u} + \frac{1}{u^3}, \qquad \left(\frac{1}{u+1}\right)^2 + \frac{1}{u+1} = \frac{u}{(u+1)^2}$$

prove the map equation. The degree of R is three, with nonzero derivative $u^2/(u+1)^2$. The two quadratic function-field extensions give $[k(C) : k(E)] = 3$, separable. The rational map extends to the smooth projective curves. At $u = 0$ it has ramification index three; the points over $u = 1$ and source infinity map unramified to O.

Explicit Jacobian transfer. Because $\pi^* O \sim 3\infty$, for $P = (x,y)$ with $x \neq 0$, the reduced pullback is

$$\boxed{U(u) = u^2 + x^{-1}u + x, \qquad V(u) = \big(x + (y+1)x^{-2}\big)u + x + y + 1.}$$

Indeed the cubic fiber is $A = u^3 + xu^2 + x$, $B = \kappa u^2 + xu + x$, $\kappa = (y + 1 + x)/x$. One Cantor reduction gives $(f + u^2 B + B^2)/A = U$, $(u^2 + B) \bmod U = V$. At O use $(1,0)$; at $(0,1)$ use $(u,0)$.

Inverse and group law. On this image, recover $x = U(0)$, $y = V(0) + x + 1$; handle O and $(0,1)$ separately. Validate and reconstruct the whole pair before accepting it. Uniqueness of reduced pairs makes this inverse injective. Picard pullback is a homomorphism. Independently, $\pi_* \pi^* = [3]$ proves subgroup injectivity whenever $\gcd(3, r) = 1$. Both requested prime orders satisfy this condition. A norm on arbitrary Jacobian classes is not implemented.

Native evidence on the requested curves

The field basis is $z^{83} + z^{45} + z^2 + z + 1$, modulus `0x80000000200000000007`. Source fixtures are pinned in `n83-final.json`. For $a = 0$ the historical Sage prime receipt is reused; for $a = 1$ deterministic u64 primality is freshly checked.

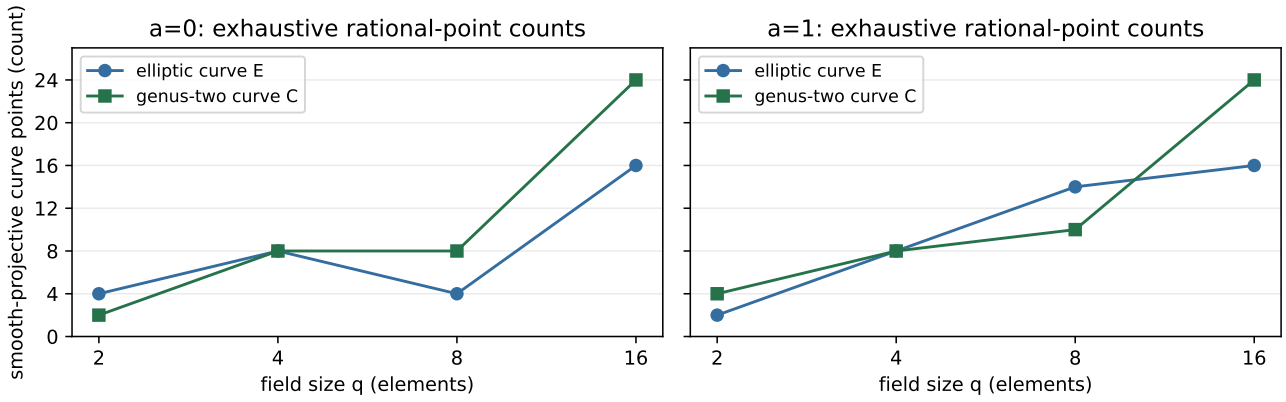
a	Prime subgroup order r	Cofactor
0	241785163923079621668568 9	4
1	8569786107849059	1128547018

For each case, native checks establish $rP = rQ = O$, valid nonzero D_P, $rD_P = rD_Q = 0$, inverse round trips, eight scalar relations, eight additions, and $D_Q = 37D_P$. Elliptic fixture arithmetic uses an independent u128 implementation; Jacobian arithmetic uses the existing BigUint Cantor backend.

One concrete constructed divisor, for the $a = 0$ generator, is:

```
U = [0x477f77103dfad59850800, 0x34dcc667b797c42263667, 0x1]
V = [0x68da9027e8b8117caddc2, 0x14128f4ae40303ac38b16]
```

Coefficients are in ascending powers. Its order is exactly the first row's r, using the cited prime receipt, nonidentity and $rD_P = 0$.



run-20261014T1442Z · n83-final.json small_field_controls · complete counts; no sampling uncertainty

All rational source-point maps and every ordered elliptic addition pair over fields of sizes $2, 4, 8, 16$ passed for both a values. These are exhaustive curve-point counts, not Jacobian orders. The plot displays recorded values only. No field-wide scan at 2^{83} was attempted or required.

Negative controls reject invalid points, fields, widths and divisors; a valid divisor outside the image; malformed CLI input; and a wrong scalar witness. The IDC1 checker also rejects a false identity and a tampered digest.

Proof scope and reproduction

Checked Lean theorem. Lean 4.24.0, commit `797c613eb9b6d4ec95db23e3e00af9ac6657f24b`, with no external dependencies, checks the actual characteristic-two affine map equation for all field elements satisfying the source equation and $u + 1 \neq 0$.

Declaration	Scope
`cleared_map_identity`	Cleared polynomial equality over every characteristic-two field.
`map_affine_point`	Actual inverse-denominator coordinates satisfy the elliptic equation.
`image_constants_recover_point`	Constant-coordinate recovery equality.

`lake build` and the direct axiom audit pass, reporting only `propext`, `Classical.choice`, `Quot.sound`. No `sorryAx` occurs in the checked declarations. `lean4checker` was not installed.

Integration status. Scoped package checks pass in both repositories. The full `crypto` library suite has 4087 passes, two existing JV F4 minimum-completion failures and 117 ignored tests. Both failed tests pass serially on the same binary. Its library source is unchanged; the full-suite failure and unresolved cause are retained in the run's validation record.

Partial formalization. Genus, projective completion, exceptional fibers, divisor-class equivalence, Cantor correctness and Picard functoriality have the written proof on page 2 and native checks. They have not been formalized as actual geometric Lean objects. The report's proof map and named problem retain these obligations. The algebraic theorem is not represented as a full formal Jacobian proof.

Identity certificate. `IDC1h7cbc5bff87f8b59d` checks the exact integer lift with an explicit even correction. Native coefficient expansion agrees exactly; 32 certificate points and 32 additional checker points pass over $2^{61} - 1$. Reduction modulo two supplies the characteristic-two identity. This is same-host evidence.

Commands from either repository root:

```
cargo run --release --locked \
  --manifest-path tools/binary-genus2-cover/Cargo.toml -- \
  --a both --output /tmp/genus2-n83-final.json
tools/binary-genus2-cover/check.sh
cd research/binary_genus2_cover_20261014/lean
lake build
lean BinaryGenusTwo.lean
```

For supplied Q, choose one a, add `--q-point 0xX,0xY` and omit `--scalar` if unknown. The scalar fields then remain null.

Sources. The editable Markdown report, Rust source, protocol, frozen `n83-final.json`, exact identities, source hashes, successful validation and failed preflights are beside this PDF. Standard geometric references: Stichtenoth, *Algebraic Function Fields and Codes*, §3.7, and Milne, *Jacobian Varieties*. This construction supplies neither a descent to a smaller field nor an unknown-log solution or a faster ECDLP solver.